



From pilot to practice

Technical evidence and an executive decision

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Agenda

Decision context

Technical evidence

Executive recommendation

A synthetic pilot: evidence, tradeoffs, and a bounded next step.

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01 / Decision context

What must be true before expanding the pilot?

A smaller pilot can answer the largest uncertainty

- The team needs reliable predictions within a fixed latency budget.

Note: scope of the evidence

All results here are synthetic; they illustrate a presentation structure.

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- A controlled pilot separates model quality from workflow effects.

Note: scope of the evidence

All results here are synthetic; they illustrate a presentation structure.

A smaller pilot can answer the largest uncertainty

- The team needs reliable predictions within a fixed latency budget.
- A controlled pilot separates model quality from workflow effects.
- Expansion depends on both technical and operational evidence.

Note: scope of the evidence

All results here are synthetic; they illustrate a presentation structure.

Compare alternatives against the same constraints

Option A / baseline

Lower integration effort.

Predictable operating cost.

Smaller quality improvement.

Option B / encoder

Better pilot accuracy.

Additional monitoring work.

Fits the latency budget.

Shared conclusion: test Option B in a bounded deployment.

The workflow makes each transformation inspectable



Source: original editable TikZ diagram; synthetic workflow.

Tip: retain observability

Record latency and review outcomes at each boundary.

A simple loss connects predictions to labels

For batch size B and class count C , minimize cross-entropy:

$$\mathcal{L} = -\frac{1}{B} \sum_{i=1}^B \sum_{c=1}^C y_{ic} \log p_{ic}. \quad (1)$$

Attention-based representations motivate the encoder (Vaswani et al., 2017).

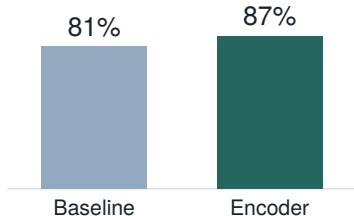
Warning: validation is separate from training

Select operating thresholds on held-out data, then test once.

Quality improves within the pilot latency budget

Model	Accuracy	Latency
Baseline	81%	12 ms
Encoder	87%	18 ms

Interpretation: +6 percentage points at +6 ms.



Source: synthetic held-out pilot, $n = 1,000$; accuracy bars start at zero. See equation 1.

Use images with a descriptive source and clear purpose



The shared T mark combines a navy crossbar and stem with a teal time tick.

Source: bundled vector artwork, exported from `\report\logomark`; original Tempus design.

Keep implementation examples short and readable

```
def accept_candidate(accuracy, latency_ms):  
    # Synthetic operating gates.  
    quality_ok = accuracy >= 0.85  
    latency_ok = latency_ms <= 20  
    return quality_ok and latency_ok
```

Note: code is optional

Enable the theme's listings option and use a fragile frame. Split long examples across frames; the default code size is 8 pt.

Three metrics describe the pilot, not the whole system

Quality

87%

Accuracy

Held-out sample

$n = 1,000$

Speed

18 ms

Median latency

Single request

CPU inference

Effort

2 weeks

Pilot duration

Two engineers

Fixed scope

Context: synthetic planning figures; no production guarantee.

Four milestones keep expansion reversible

1. Week 1 / Prepare: freeze data and success criteria.
2. Week 2 / Evaluate: verify quality and latency.
3. Week 3 / Pilot: monitor a limited user cohort.
4. Week 4 / Decide: expand, revise, or stop.

Each milestone has an owner and a recorded decision.

Approve a bounded pilot with explicit stopping criteria

Important: decision requested

Approve a two-week pilot of the compact encoder.

Evidence: quality improves by six percentage points while median latency stays below 20 ms.

Next step: assign an owner and review drift, latency, and user outcomes before expanding the cohort.

References

Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N Gomez, Łukasz Kaiser, and Illia Polosukhin. 2017. Attention is all you need. In *Advances in Neural Information Processing Systems*, volume 30. Curran Associates, Inc.

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A decision grounded in evidence

Questions and discussion

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